

Safety Verification of Open Multi-Agent Systems

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Abstract

Open Multi-Agent Systems (MAS) allow agents to join and leave dynamically, creating significant challenges for ensuring system safety. This project applies Petri nets as a formal modeling tool to capture the concurrent behavior of agents and analyze the reachability of unsafe states through coverability techniques. By constructing agent-level models, composing a global Petri net, and applying automated verification tools such as TAPAAL, PIPE, and LoLA, the project aims to determine whether safety properties hold under system evolution. Initial progress confirms the feasibility of Petri net based safety verification for open MAS, with further work focusing on global modeling and comprehensive coverability analysis.

1 Introduction

Multi-Agent Systems (MAS) are distributed systems composed of autonomous agents that interact to achieve individual or collective objectives. In open MAS, agents can dynamically join or leave the system, making the verification of safety properties critical. Ensuring that the system never reaches undesirable states such as deadlocks, resource conflicts, or unsafe interactions is the core challenge of this project.

Petri nets are used as a formalism to model the concurrent and distributed behaviors of agents. Coverability analysis of Petri nets allows us to determine whether unsafe states can be reached from the initial system configuration.

2 Objectives

- To model individual agents in an open MAS using Petri nets.
- To compose a global Petri net representing the interactions of all agents.
- To identify unsafe states and verify their coverability.
- To utilize automated tools for safety verification.

3 Methodology

1. **Agent Modeling:** Each agent is represented as a Petri net capturing its possible states and actions.
2. **System Composition:** Individual agent nets are combined to form a global model that reflects all interactions.
3. **Unsafe Markings Identification:** Critical states where system safety is violated are defined.
4. **Coverability Analysis:** Using tools like TAPAAL and PIPE, the global Petri net is analyzed to check whether unsafe markings are reachable.
5. **Result Interpretation:** Based on coverability analysis, the system's safety is determined.

4 Tools and Environment

- **TAPAAL:** For modeling timed Petri nets and performing coverability analysis.
- **PIPE:** Platform Independent Petri Net Editor used for simulation and state-space exploration.
- **LoLA:** Low-Level Petri Net Analyzer for formal verification.

5 Progress Till Date

- Literature review completed on MAS, open MAS, and Petri net modeling.
- Identification of key safety properties and unsafe states.
- Preliminary Petri net models for individual agents developed.
- Setup of TAPAAL and PIPE environments for simulation and analysis.
- Conducted initial coverability tests on simplified agent interactions.

6 Next Steps

- Complete the global Petri net model with multiple interacting agents.
- Perform comprehensive coverability analysis for all identified unsafe markings.
- Document results with diagrams and examples.
- Explore potential extensions such as timed and colored Petri nets.
- Prepare final report and presentation.

7 Conclusion

The project is progressing as planned. Initial modeling and tool setup have been completed, and preliminary analysis demonstrates the feasibility of verifying safety in open MAS using Petri nets. The next phase will focus on global system modeling and detailed coverability verification to ensure robustness and safety of the system.